

Fertilization at planting influences seedling growth and vegetative competition on a post-mining boreal reclamation site

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Abstract Reclamation of post-mining sites is challenged by limiting factors including adverse soil chemical and physical properties, along with weed competition. Fertilization may alleviate nutrient deficiencies, but broadcast fertilization with immediately available fertilizers (IAF) results in generally low rates of nutrient recovery for planted trees. Directed application of controlled-release fertilizer (CRF) to the rhizosphere offers an alternative to extend nutrient longevity while reducing nutrient leaching or uptake by competing vegetation. We evaluated white spruce (*Picea glauca* (Moench) Voss) and aspen (*Populus tremuloides* Michx.) development on a mine reclamation site in the Oil Sands Region of northern Alberta in response to IAF and polymer-coated CRF (3–4 and 8–9 month release). IAF and CRF (each NPK plus other macro- and micro-nutrients) were applied at 20 and 40 g N seedling⁻¹ and 2 and 4 g N seedling⁻¹, respectively. No effect on seedling survival occurred. White spruce year-1 height and diameter growth responded positively to both IAF and CRF relative to non-fertilized controls, but in year-2 seedlings treated with CRF (i.e., 8–9 month) outperformed those fertilized with IAF, which were similar to controls. Aspen height growth did not respond strongly to fertilization in either year, but diameter growth showed positive responses to all fertilizer treatments in year-1 and to all CRF treatments and the high IAF rate in year-2. Responses were attributed to a pronounced increase in vegetative competition observed for IAF relative to CRF or controls following the first growing season, and generally higher levels of foliar N in year-1 for CRF compared to IAF or unfertilized trees. Thus, directed root zone application of CRF promoted seedling growth responses similar to or better than those induced by broadcast IAF applications, but at 90–95 % lower N application rates.

Keywords Competitive interactions · Controlled-release fertilizer · Forest restoration · Mine reclamation · Nutrient management · Outplanting

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Introduction

Forest restoration programs such as mine reclamation are important to the maintenance and re-establishment of ecosystem services on disturbed and degraded sites (Ciccarese et al. 2012; Oliet and Jacobs 2012). The complexity of restoration strategies employed on such sites following industrial disturbance has been shown to influence the trajectory of ecosystem recovery (Rayfield et al. 2005; MacDonald et al. 2012). Rapid establishment of forest tree seedlings following mining is necessary to ensure effective restoration to forests, but these sites can be difficult to reforest. Low soil fertility, excessively high or low soil pH, salinity, soil compaction, competition from weeds, and animal browse are key factors that may limit early establishment success of newly outplanted seedlings on mine reclamation sites (Bussler et al. 1984; Andersen et al. 1989; Casselman et al. 2006). These biotic and abiotic stresses impair basic plant physiological processes such as CO₂ assimilation, respiration, translocation, and synthesis of sugars, proteins, and nucleic acids, which may restrict the ability of seedlings to establish on a site and lead to slow growth and/or mortality (Salifu et al. 2009; Grossnickle 2012). Thus, a need exists to identify cost-effective and practical silvicultural approaches to promote successful forestland restoration on mine reclamation sites.

Attempts to improve regeneration success on such sites have included herbicide use and field fertilization (Andersen et al. 1989; Casselman et al. 2006). Although herbicide application may effectively control competing vegetation and improve seedling establishment success (Andersen et al. 1989), there is increasing public sentiment against its use owing to negative effects on the environment and on biodiversity (Thiffault and Roy 2011; Löf et al. 2012). Field fertilization may promote seedling establishment and recovery of ecosystem function on restored sites (Rowland et al. 2009). However, broadcast fertilization with traditional (agronomic) immediately available fertilizers (IAF) releases nutrients immediately upon application with generally low rates of fertilizer recovery. For example, broadcast field fertilization has been shown to stimulate growth and nutrient uptake of competing vegetation more than outplanted seedlings (Chang et al. 1996; Staples et al. 1999; Chang and Preston 2000; Imo and Timmer 1998; Ramsey et al. 2001). Additionally, the rapid release of nutrients immediately following application of IAF creates potential for environmental contamination through leaching.

In contrast to conventional fertilization techniques, controlled-release fertilizers (CRF) offer an alternative that may provide plants with enhanced mineral nutrition for extended periods. CRF differ from conventional forms of fertilizer (e.g., urea and water-soluble products) in that the majority of nutrients are not available immediately following application, but are released slowly over time, with release rates ranging from about 3–18 months. Thus, a single application of CRF can supply seedlings with enhanced nutrition for up to 2 years (Jacobs et al. 2005), providing a more consistent and sustained flow of nutrients that better coincides with plant development and seedling nutrient demand (Donald 1991). Additionally, CRF may be applied directly (or immediately adjacent) to the seedling root zone at planting with relatively low risk of root damage compared to IAF (Jacobs and Timmer 2005). This improves overall fertilizer use efficiency and helps to minimize nutrient leaching (Hangs et al. 2003).

CRF primarily vary in terms of their nutrient formulations, estimated product longevities, and mechanisms of nutrient release. Polymer-coated CRF are considered the most technologically advanced form of CRF due to the considerable ability to control product longevity and subsequent efficiency of nutrient delivery. Nutrient release of most polymer-coated CRF is determined by the diffusion of water through the semi-permeable membrane

(Goertz 1993), a process that is accelerated at progressively higher soil temperatures with soil water content having relatively little influence on release (Kochba et al. 1990). Application of CRF at planting has stimulated growth of newly planted forest trees across a variety of ecotypes in North America (e.g., Carlson 1981; Carlson and Preisig 1981; Arnott and Burdett 1988; Fan et al. 2002; Jacobs et al. 2005). Despite the demonstrated potential to improve efficiency of fertilizer operations on reforestation and afforestation sites, the relative effectiveness of CRF versus immediately available fertilizer forms remains largely untested on (inherently) heavily disturbed mine reclamation sites. Additionally, most studies investigating use of CRF in field plantings have not performed direct comparisons between CRF and IAF (and their corresponding application methods) in terms of seedling development or influences on competing vegetation.

In the Oil Sands Region of northern Alberta, Canada, large areas of land have been disturbed by open-pit surface mining (currently estimated at 714 km², with 71 km² under active reclamation; Government of Alberta 2013). Environmental law requires that these areas be reclaimed to “equivalent land capability” after industrial mining operations, which involves re-contouring of the land, soil reconstruction using a mix of soils salvaged from existing boreal forests and peatlands, and (often) planting of native boreal forest trees. These reclamation sites are frequently treated with broadcast application of IAF for up to 5 years (Pinno et al. 2012). However, detection of excessively high or low nitrate levels (Rowland et al. 2009) in some fertilized stands in the Oil Sands Region suggests the need to optimize fertilization operations to improve cost efficiency and decrease the risk of nutrient leaching to surface and underground water bodies (McMillan et al. 2007).

To help address these knowledge gaps, we initiated a field trial on a mine reclamation site in the Canadian Oil Sands Region to examine use of CRF in comparison to IAF. Our objectives were to (1) characterize seedling growth and nutrient uptake over the course of two growing seasons in response to varying types and application rates of CRF vs. IAF, and (2) examine response of competing vegetation to the varying fertilizer treatments and their application methods (i.e., planting hole fertilization for CRF and broadcast for IAF).

Materials and methods

Study site and plantation establishment

On 9 June 2011, two experiments were planted into a peat-mineral mix capping material with an average depth of 50 cm overlaying an overburden substrate on a level portion of the MD8 reclamation area of the Suncor Energy mine north of Fort McMurray, Alberta, Canada (56°56.495'N, 111°16.443'W, 369 m ASL) to evaluate the effects of IAF and CRF applied at time of planting on white spruce (*Picea glauca* (Moench) Voss) and aspen (*Populus tremuloides* Michx.) seedling performance in a post-mining boreal forest reclamation context. These two species, along with jack pine (*Pinus banksiana* Lamb.), represent the common forest tree species used for reclamation plantings in the Oil Sands Region in northeastern Alberta (Sorenson et al. 2011). Mean historical summer/winter (May–August/September–April) daily air temperature and seasonal precipitation in Fort McMurray are 14.3/–6.0 °C and 265.7/190.0 mm, respectively (Environment Canada: National Climate Data and Information Archive 2013).

Independent monocultural planting experiments were established for each species. Seedlings of both species were grown in 615A styroblock containers (336 cm³ volume, 15.1 cm depth × 5.9 cm top diameter, Beaver Plastics, Ltd., Acheson, Alberta, Canada)

and operationally produced by Coast to Coast Reforestation Inc. at the Smoky Lake Forest Nursery (Smoky Lake, Alberta, Canada). For each experiment, a randomized complete block design was employed consisting of four replicate blocks. Each block contained 84 experimental seedlings hand-planted at a spacing of 2 m × 2 m (equivalent to 2,500 seedlings ha⁻¹), with each block surrounded by a buffer row of non-experimental seedlings planted at the same density. At time of planting, seedlings were assigned to one of seven experimental fertilizer treatments (described below).

Fertilization

The day after planting, each experimental seedling received one of seven fertilization applications; buffer row seedlings received no fertilizer. Fertilizer treatments were as follows: unfertilized control (abbreviated UC); 100 g seedling⁻¹ immediately available 20N–20P–20K (i.e., 20 g N seedling⁻¹, equivalent to 200 kg N ha⁻¹; abbreviated IA-L) Peter's Professional (Scott's MiracleGro, Marysville, OH, USA); 200 g seedling⁻¹ immediately available 20N–20P–20K (i.e., 40 g N seedling⁻¹, equivalent to 400 kg N ha⁻¹; abbreviated IA-H) Peter's Professional; 13.3 g seedling⁻¹ 3–4 month release 15N–9P–12K (i.e., 2 g N seedling⁻¹; abbreviated FR-L) controlled-release Osmocote (Scott's MiracleGro, Marysville, OH, USA); 26.7 g seedling⁻¹ 3–4 month release 15N–9P–12K (i.e., 4 g N seedling⁻¹; abbreviated FR-H) controlled-release Osmocote; 13.3 g seedling⁻¹ 8–9 month release 15N–9P–12K (i.e., 2 g N seedling⁻¹; abbreviated SR-L) controlled-release Osmocote (Scott's MiracleGro, Marysville, OH, USA); or 26.7 g seedling⁻¹ 8–9 month release 15N–9P–12K (i.e., 4 g N seedling⁻¹; abbreviated SR-H) controlled-release Osmocote (Table 1). IAF were applied by surface broadcasting the prescribed amount evenly within a 1-m² frame centered on the target seedling. CRF were applied by dibbling the prescribed amount into a hole at 15 cm depth in the soil 5–7 cm uphill from the target seedling, a method that has shown effective fertilizer delivery in past trials (Hangs et al. 2003; Hawkins et al. 2005).

Measurements

Seedling survival, total height to the base of the top-most bud (cm), root collar diameter (mm), and competing vegetation cover (% cover within a 1-m sampling grid centered on each seedling) were measured on each seedling at time of planting in June 2011 and at the

Table 1 Fertilizer treatment characteristics

Treatment	Product type	Labeled release rate	Nutrient ratio	Product application rate (g seedling ⁻¹)	Amount of N applied (g N seedling ⁻¹)
UC	N/A	N/A	N/A	0	0
IA-L	IAF	N/A	20N–20P–20K	100	20
IA-H	IAF	N/A	20N–20P–20K	200	40
FR-L	CRF	3–4 months	15N–9P–12K	13.3	2
FR-H	CRF	3–4 months	15N–9P–12K	26.7	4
SR-L	CRF	8–9 months	15N–9P–12K	13.3	2
SR-H	CRF	8–9 months	15N–9P–12K	26.7	4

N/A not applicable, IAF immediately available fertilizer, CRF controlled-release fertilizer

end of the first and second growing seasons in September 2011 and October 2012, respectively. Relative height growth for each seedling was calculated as the change in height during a season divided by the initial height at the beginning of that season, with the result being expressed as a percentage. Relative root collar diameter growth was calculated similarly. In September 2011, foliar tissue sub-samples were collected from the upper one-third of each seedling and combined to create a composite foliar tissue sample for each treatment in each block; in October 2012, foliar tissue was again collected as described above, but only from the white spruce experiment due to senescence of aspen leaves at the time of collection. Following collection, foliar tissue samples were air-dried to constant mass, ground to pass a 20-mesh screen, and submitted to A & L Great Lakes Laboratories (Fort Wayne IN, USA) for nutrient analyses. Plant N was determined by combustion (“Dumas”) procedure (AOAC 968.06) using a LECO nitrogen analyzer (LECO Corp., St. Joseph, MI, USA). Additionally, plant samples were digested in nitric + perchloric acids (AOAC 935.13), and the other elements determined using inductively coupled argon plasma (ICAP) analysis (AOAC 985.01).

Statistical analyses

White spruce and aspen data were analyzed separately, with each species constituting a separate experiment. Each fertilizer product and rate combination was treated as a single level of the fertilizer treatment main effect. Survival, seedling absolute height and diameter, seedling relative height and relative root collar diameter growth, and vegetative competition data were analyzed as a randomized complete block design using repeated measures ANOVA to detect main effect significance ($P < 0.05$) followed by Tukey’s multiple pairwise comparison to identify significant differences between fertilizer treatments within a given growing season ($\alpha = 0.05$). Foliar nutrition data were analyzed within each species as a randomized complete block design using repeated measures ANOVA for white spruce data and one-way ANOVA for aspen data to detect main effect significance ($P < 0.05$) followed by Tukey’s multiple pairwise comparison to detect significant differences between treatments within a species and growing season ($\alpha = 0.05$).

Results

White spruce seedling survival, growth, and foliar nutrition in response to fertilization

Averaged across all treatments, white spruce seedling survival was 99.4 % after two growing seasons and was not significantly influenced by fertilizer treatment during the experiment ($P = 0.5448$, data not shown).

Both absolute height and relative height growth of white spruce were significantly affected by fertilization ($P = 0.0004$ and $P < 0.0001$, respectively; Fig. 1). During both the first and second growing seasons, absolute height was positively influenced by all fertilization regimes except the IA-L treatment relative to unfertilized controls (Fig. 1). For relative height growth during the first growing season, all treatments showed greater relative height growth compared to UC seedlings except SR-H, while during the second season, SR-L out-performed FR-H, IA-L, and UC (Fig. 1).

White spruce absolute root collar diameter and relative root collar diameter growth also responded to fertilization ($P < 0.0001$ and $P = 0.0006$, respectively; Fig. 2), with all fertilized treatments showing both greater root collar diameters and relative root collar

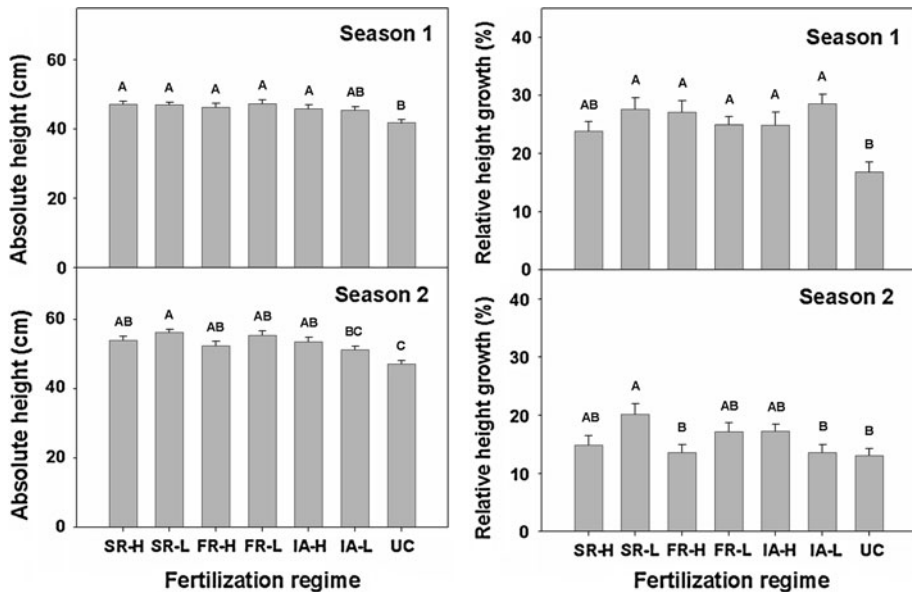


Fig. 1 White spruce absolute height (*left*) and relative height growth (*right*). Values displayed are means ($\pm$ SE). Means marked with the same letter within a given season did not differ significantly ($\alpha = 0.05$). *SR-H* slow-release (8–9 month release) controlled-release fertilizer (CRF) at high (4 g N seedling⁻¹) rate, *SR-L* slow-release CRF at low (2 g N seedling⁻¹) rate, *FR-H* fast-release (3–4 month) CRF at high (4 g N seedling⁻¹) rate, *FR-L* fast-release CRF at low (2 g N seedling⁻¹) rate, *IA-H* immediately available fertilizer (IAF) at high (40 g N seedling⁻¹) rate, *IA-L* IAF at low (20 g N seedling⁻¹) rate, *UC* unfertilized control

diameter growth after the first growing season. During the second growing season, all CRF treatments showed increased root collar diameter size relative to UC seedlings (Fig. 2), and the SR-L and UC treatments both had greater relative root collar diameter growth compared to the IA-H and IA-L seedlings (Fig. 2).

Fertilization treatments significantly affected foliar concentrations of N, P, and K in white spruce seedlings ($P < 0.0001$ for N and P, and $P = 0.0155$ for K; Table 2). Following the first growing season, all CRF treatments and the IA-H treatment showed elevated foliar N concentrations relative to the IA-L and UC treatments (Table 2); however, no significant differences in foliar N concentrations were observed at the end of the second growing season. Foliar P concentrations were elevated in the IA-H treatment relative to other treatments following the first growing season, while both the IA-H and IA-L treatments showed elevated foliar P concentrations relative to SR-L and FR-L seedlings following the second growing season (Table 2). Foliar K concentrations did not differ between treatments after the first growing season, while IA-L had greater foliar K levels than SR-L seedlings after the second growing season (Table 2).

Aspen seedling survival, growth, and foliar nutrition in response to fertilization

Overall aspen seedling survival was 81.2 % after two growing seasons, but survival was not significantly affected by fertilizer treatment during this period ($P = 0.1099$, data not shown).

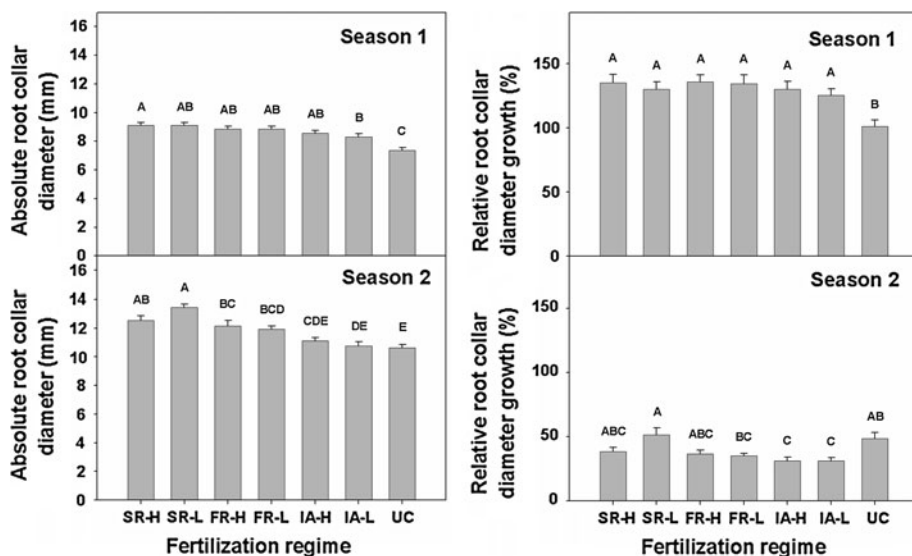


Fig. 2 White spruce absolute root collar diameter (*left*) and relative root collar diameter growth (*right*). Values displayed are means ($\pm$ SE). Means marked with the same letter within a given season did not differ significantly ($\alpha = 0.05$). Abbreviations as in Fig. 1

Absolute height of aspen was significantly affected by fertilization treatment ($P = 0.0006$; Fig. 3), with seedlings of the SR-L, IA-H, and IA-L treatments exhibiting greater absolute heights at the end of the first growing season than those of the unfertilized control group; however, no differences in absolute height were detected between treatments during the second growing season (Fig. 3). Aspen relative height growth responded in a manner similar to absolute height ($P = 0.0112$; Fig. 3).

Absolute root collar diameter and relative root collar diameter of aspen seedlings were both strongly influenced by fertilization treatments ($P < 0.0001$ for both; Fig. 4). During the first growing season, all fertilized seedlings out-performed unfertilized control seedlings for both parameters. During the second growing season, the SR-H, SR-L, and FR-H treatments exhibited larger absolute root collar diameters relative to the IA-L or unfertilized control seedlings, while SR-H, FR-H, and FR-L seedlings out-performed IA-L seedlings with regard to relative root collar diameter growth (Fig. 4).

Fertilization treatments significantly influenced foliar concentrations of N, P, and K in aspen seedlings ($P < 0.0001$, $P = 0.0293$, and $P = 0.0002$ for N, P, and K, respectively; Table 2). All CRF treatments and the IA-H treatment had higher foliar N concentrations after the first growing season, with the SR-H and FR-H treatments having higher foliar N concentrations than IA-H seedlings (Table 2). Foliar P concentrations were lower in IA-L seedlings compared to UC seedlings, while foliar K concentrations increased in SR-H, FR-H, FR-L, and IA-H seedlings relative to UC seedlings (Table 2).

Response of vegetative competition to fertilizer treatments

In white spruce plots, cover of vegetative competition was significantly influenced by fertilization treatment ($P < 0.0001$; Figs. 5, 6). Both the IA-H and IA-L treatments resulted in substantial increases in competing vegetation cover following the first growing

Table 2 Response of foliar nutrient concentrations to fertilization treatments

Species	Fertilizer treatment	Season 1			Season 2		
		N (%)	P (%)	K (%)	N (%)	P (%)	K (%)
Aspen	SR-H	3.30 (0.11) A	0.15 (0.01) AB	1.10 (0.07) A	–	–	–
	SR-L	2.98 (0.05) BC	0.15 (0.00) AB	0.99 (0.04) ABC	–	–	–
	FR-H	3.22 (0.09) AB	0.16 (0.01) AB	1.08 (0.02) AB	–	–	–
	FR-L	3.12 (0.08) ABC	0.15 (0.00) AB	1.12 (0.02) AB	–	–	–
	IA-H	2.87 (0.04) C	0.16 (0.00) AB	1.10 (0.06) A	–	–	–
	IA-L	2.36 (0.04) D	0.14 (0.00) B	0.88 (0.04) BC	–	–	–
	UC	2.21 (0.03) D	0.17 (0.01) A	0.79 (0.04) C	–	–	–
White spruce	SR-H	1.72 (0.05) A	0.08 (0.00) B	0.54 (0.02) A	1.21 (0.06) A	0.13 (0.01) ABC	0.59 (0.02) AB
	SR-L	1.52 (0.11) AB	0.08 (0.00) B	0.55 (0.01) A	0.91 (0.07) A	0.11 (0.01) C	0.55 (0.02) B
	FR-H	1.72 (0.09) A	0.08 (0.00) B	0.61 (0.02) A	1.22 (0.08) A	0.12 (0.01) BC	0.60 (0.02) AB
	FR-L	1.51 (0.04) AB	0.08 (0.00) B	0.53 (0.02) A	1.00 (0.07) A	0.11 (0.01) C	0.59 (0.03) AB
	IA-H	1.37 (0.04) B	0.10 (0.00) A	0.60 (0.03) A	1.01 (0.07) A	0.17 (0.01) A	0.64 (0.01) AB
	IA-L	1.03 (0.05) C	0.08 (0.00) B	0.56 (0.03) A	0.93 (0.05) A	0.16 (0.00) AB	0.68 (0.02) A
	UC	0.85 (0.05) C	0.08 (0.00) B	0.60 (0.01) A	1.21 (0.11) A	0.14 (0.01) ABC	0.63 (0.02) AB

Values displayed are mean percentages ($\pm$ SE) based on dry mass. Means marked with the same letter for a given species and nutrient did not differ significantly ($\alpha = 0.05$) *SR-H* Slow-release (8–9 month release) controlled-release fertilizer (CRF) at high (4 g N seedling⁻¹) rate, *SR-L* slow-release CRF at low (2 g N seedling⁻¹) rate, *FR-H* fast-release (3–4 month) CRF at high (4 g N seedling⁻¹) rate, *FR-L* fast-release CRF at low (2 g N seedling⁻¹) rate, *IA-H* immediately available fertilizer (IAF) at high (40 g N seedling⁻¹) rate, *IA-L* IAF at low (20 g N seedling⁻¹) rate, *UC* unfertilized control

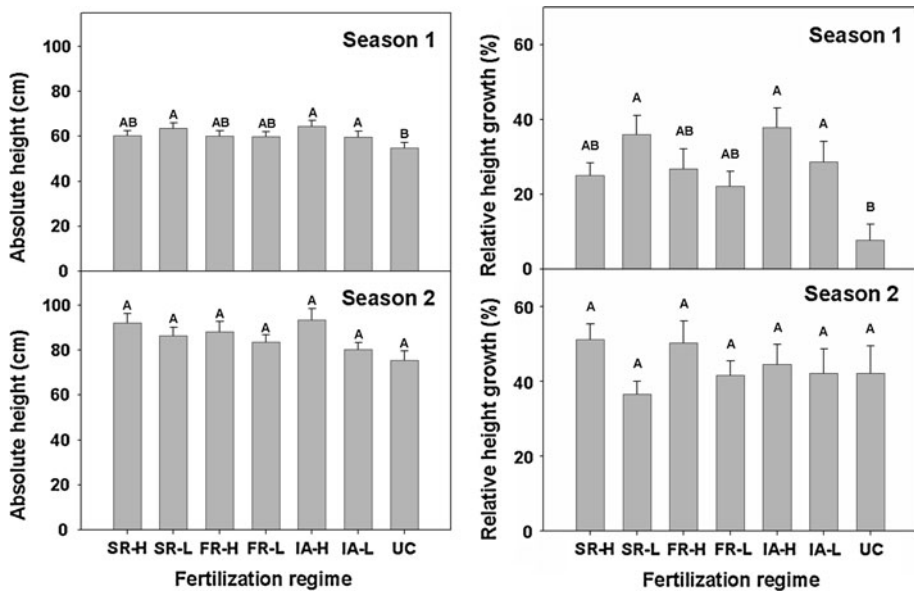


Fig. 3 Aspen relative height growth. Values displayed are means ($\pm$ SE). Means marked with the same letter within a given season did not differ significantly ($\alpha = 0.05$). Abbreviations as in Fig. 1

season, which persisted into the second growing season. Although none of the CRF treatments showed increased vegetative competition relative to UC seedlings during the first growing season, all had elevated levels of competition relative to controls following the second season (Fig. 5).

Vegetative competition in aspen was also significantly influenced by fertilization treatments ($P = 0.0064$; Fig. 5), with the IA-H and IA-L treatments exhibiting much higher levels of competition compared to any other treatment during the first growing season, while CRF treatments showed no difference in vegetative competition relative to the unfertilized control seedlings. Following the second growing season, only the IA-H treatment had elevated levels of competition relative to unfertilized controls.

Discussion

Fertilization did not affect survival for either white spruce or aspen seedlings during the 2 years following planting, and both species showed positive growth responses to most fertilization treatments relative to unfertilized controls during the first growing season (Figs. 1, 2, 3, 4). The absence of any negative effects of fertilization on seedling survival or growth indicates that all fertilizer product and application rate combinations examined in this study resulted in soil solution nutrient ion concentrations and electrical conductivity levels outside the range of toxicity (Jacobs and Timmer 2005). In contrast, Casselman et al. (2006) attributed poor seedling survival of several species fertilized with IAF on an Ohio mine reclamation site to moisture stress associated with fertilizer-induced salinity. Similarly, van den Driessche et al. (2003) reported a 17 % decrease in survival of fertilized aspen compared to controls when irrigation was not used. In a series of studies involving several species across five sites in the Pacific Northwest USA, planting-hole fertilization

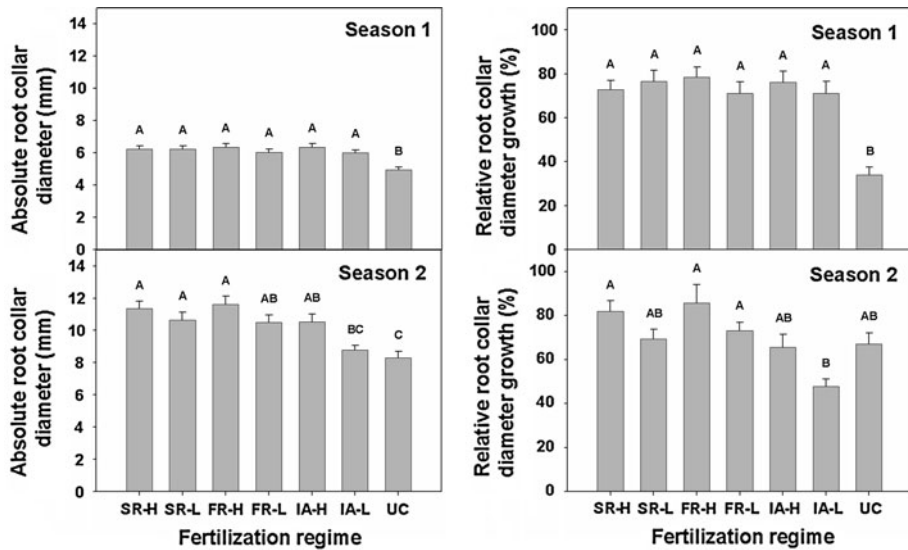


Fig. 4 Aspen absolute root collar diameter (*left*) and relative root collar diameter growth (*right*). Values displayed are means ($\pm$ SE). Means marked with the same letter within a given season did not differ significantly ($\alpha = 0.05$). Abbreviations as in Fig. 1

with two different combined formulations of IBDU (isobutylidene diurea) CRF at 3.9 g N seedling⁻¹ increased seedling stem volume only at the two sites with adequate soil moisture (Rose and Ketchum 2002), likely a function of root injury in the zone of fertilizer placement on drier sites (Jacobs et al. 2003, 2004).

During the first growing season in our study on a heavily disturbed mining site in the boreal zone, both species showed positive growth responses of similar magnitude across all fertilization treatments, with seedlings receiving 2 g N seedling⁻¹ applied as slow-release (i.e., 8–9-month release duration) CRF exhibiting height and diameter growth on par with those that received 40 g N seedling⁻¹ applied as IAF (Figs. 1, 2, 3, 4). Thus, CRF products were able to promote first season seedling growth responses at least similar to those induced through broadcast applications of IAF, but at 90–95 % lower N application rates. Furthermore, both species had generally higher mean foliar N concentrations during the first growing season when treated with CRF vs. IAF (Table 2). It is likely that this response is at least partly associated with the more directed fertilizer delivery to the target trees through application of CRF immediately adjacent to the seedling root zone versus broadcast surface application of IAF. As evidence for this effect, application of CRF 5–7 cm from roots resulted in substantially greater fertilizer use efficiency than broadcast application of CRF (Hangs et al. 2003).

Additionally, broadcast field fertilization with IAF may inadvertently increase growth of herbaceous species rather than target trees, as shown on mine reclamation sites seeded with fescue (Casselman et al. 2006). In our study, application of IAF also dramatically increased vegetative competition surrounding the seedlings of both species during the first growing season relative to all other treatments (Figs. 5, 6). Seedlings that were fertilized with CRF, however, exhibited levels of vegetative competition cover during the first growing season that did not differ from that of unfertilized controls (Figs. 5, 6). Thus, directed application of CRF adjacent to the root zone provided supplemental nutrition to seedlings during the

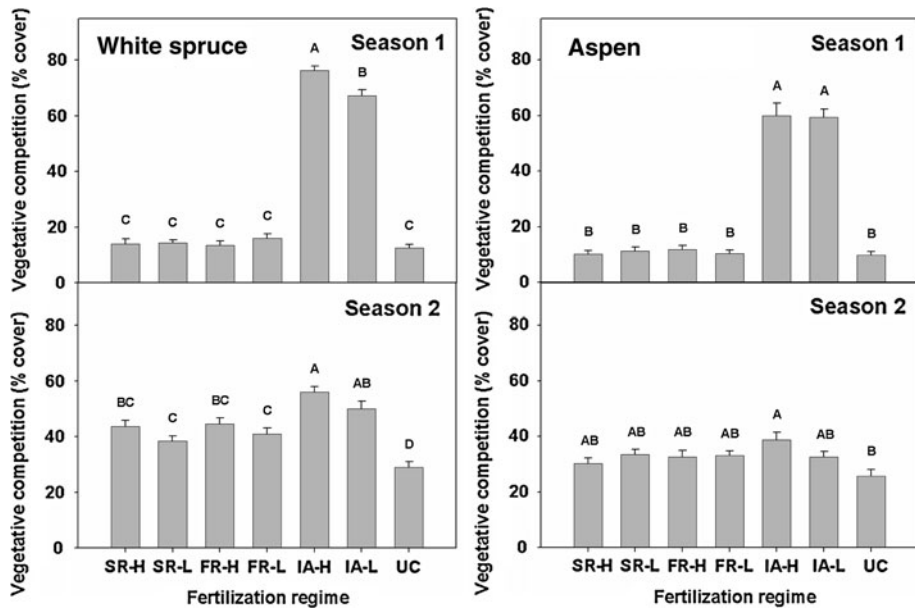


Fig. 5 Cover of vegetative competition for white spruce (*left*) and aspen (*right*). Values displayed are means ($\pm$ SE). Means marked with the same letter within a given season did not differ significantly ($\alpha = 0.05$). Abbreviations as in Fig. 1

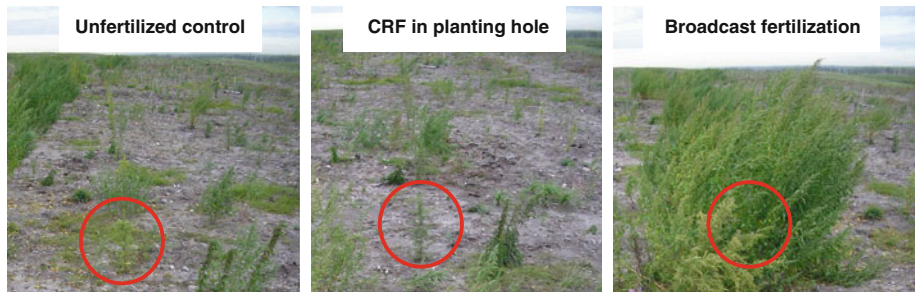


Fig. 6 Responses of competing vegetation to varying fertilization methods for white spruce plots during the first growing season. From left to right: unfertilized control; planting hole application of controlled-release fertilizer (CRF), and broadcast application of conventional (immediately available) fertilizer. Circles denote planted trees

first growing season without promoting increased levels of vegetative competition likely to increase competition for soil moisture and nutrients (Imo and Timmer 1998; Ramsey et al. 2001).

Second season growth responses varied more by species than those in the first season. White spruce seedlings continued to exhibit greater absolute heights for all CRF treatments (and IA-H) compared to controls, but only the SR-L treatment showed greater relative height growth than the control seedlings (Fig. 1). Similarly, second season absolute root-collar diameter was significantly greater for all CRF treatments compared to the control (with SR-L again having highest mean values), yet neither IAF differed from the control

(Fig. 2). This response likely reflects the gradual and sustained availability of nutrients over time associated with nutrient release patterns of CRF. Fan et al. (2002) reported that during the first growing season on a site in northern Idaho, USA, a range of 39–86 % of total nutrients released from four CRF types that varied in estimated release duration from 9 to 15 months. Jacobs et al. (2005) found that about half of the nutrients from a polymer-coated CRF were released during the first season following application and the remainder during the second year on an afforestation site in southern Indiana, USA. Although the CRF in Jacobs et al. (2005) was labeled as a 16–18 month release rate, under the longer and colder winters of northeastern Alberta we expected that our FR (3–4 month) and SR (8–9 month) CRF would release nutrients over one and two seasons, respectively. Interestingly, however, the unfertilized control showed higher second season relative root collar diameter growth than either IAF treatment, though not differing from any CRF treatment (Fig. 2). This response was likely a function of increased vegetative competition in the IAF treatments compared to controls or CRF treatments (Figs. 5, 6).

For aspen, no differences were observed between treatments for either absolute height or relative height growth during the second growing season (Fig. 3), suggesting that height growth benefits of fertilization observed for aspen during the first growing season were short-lived. Notably, however, the variation in height responses was much greater for aspen compared to white spruce (Figs. 1, 3). Aspen has been characterized as an ideal species for mine reclamation in the boreal zone based on its ability to grow over a wide range of climatic and soil conditions (Macdonald et al. 2012; Pinno et al. 2012). Nevertheless, aspen seedlings often suffer from transplant shock and exhibit several years of slow growth after outplanting (van den Driessche et al. 2003; Martens et al. 2007). This effect is likely a function of nursery stock quality associated with sub-optimal root carbohydrate reserves and root/shoot balance (Landhäusser et al. 2012), and so this tendency may have interacted to increase variability for height responses in our study. Additionally, aspen has been shown to vary in growth responses to fertilizer. For example, fertilizer only increased aspen seedling growth on a planting site in Alberta when combined with irrigation, with main responses to P fertilizer (van den Driessche et al. 2003). However, first-season trends of improved absolute diameter growth in response to fertilization largely carried over into the second growing season, with the exception that IA-L no longer differed from unfertilized control seedlings (Fig. 4). The only differences for relative root collar diameter growth were between IA-L and most CRF treatments (Fig. 4), likely a function of continued availability of CRF and interactions with vegetative competition (Fig. 5).

Our results from this study on a boreal mine reclamation site further support the contention that CRF application to the outplanting hole (or alternatively, incorporation within container root plugs) may provide a more effective means of stimulating seedling growth compared to broadcast application of IAF (Carlson and Preisig 1981; Jacobs et al. 2005; Haase et al. 2006). However, this study was conducted on a single mine reclamation site with a specific soil type (peat and mineral mix) and so results should be interpreted with caution when extrapolating to other mine reclamation sites across the landscape. The absence of observed toxicity in this experiment allows for the possibility that fertilizer rates in this experiment may have been below optimal to maximize plant growth and nutrient uptake on this reclamation site; consequently, closer examination of seedling growth responses to each product across a broader range of application rates is warranted as a prerequisite to the development of scientifically based fertilizer prescriptions for oil sands reclamation. Future research should also be designed to examine responses of CRF and broadcast fertilization to a wider range of soil types used as surface materials in mine

reclamation as fertilizer responses can vary significantly across these different soil materials (Pinno et al. 2012). Finally, the lack of differences in seedling survival among treatments, relatively modest gains in growth from fertilization (especially for aspen), and lowest incidence of competing vegetation in control (unfertilized) plots for our study suggests that refraining from fertilization could be a feasible prescription in some circumstances.

Conclusions

This study has helped to confirm the potential applicability of improved fertilizer technology (i.e., CRF) to reclamation of heavily disturbed mining sites in the Oil Sands Region of northeastern Alberta, Canada. CRF promoted first season seedling growth and nutrient uptake with responses similar to (or better than) that induced through broadcast applications of IAF, but at 90–95 % lower N application rates. Responses were generally maintained during the second growing season (though more variable for aspen height) with the slower release (i.e., 8–9 month) CRF showing generally the best responses. These effects are likely associated with three primary mechanisms: (1) directed application of CRF adjacent to the root zone versus broadcast application of IAF, (2) the gradual flow of fertilizer nutrients into the rhizosphere of planted seedlings in the CRF treatments that likely extended to the second growing season in the slower release CRF, and (3) the lack of an increase in vegetative competition in the CRF treatments compared to controls.

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